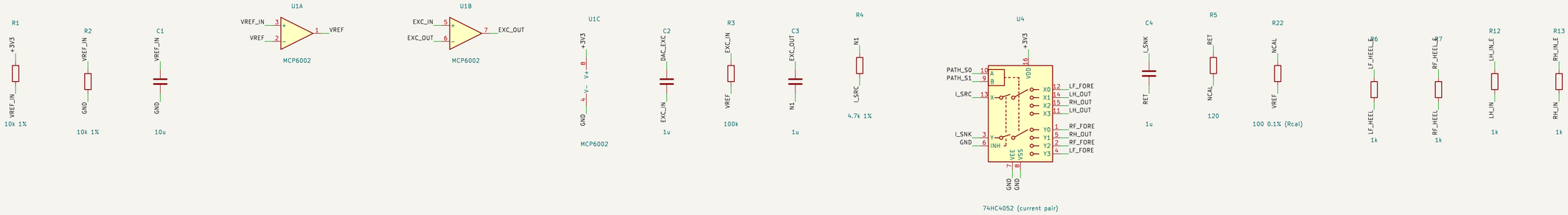
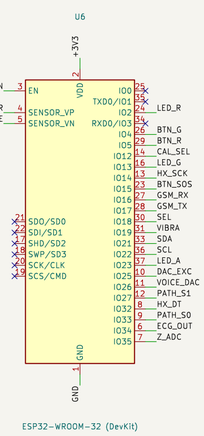
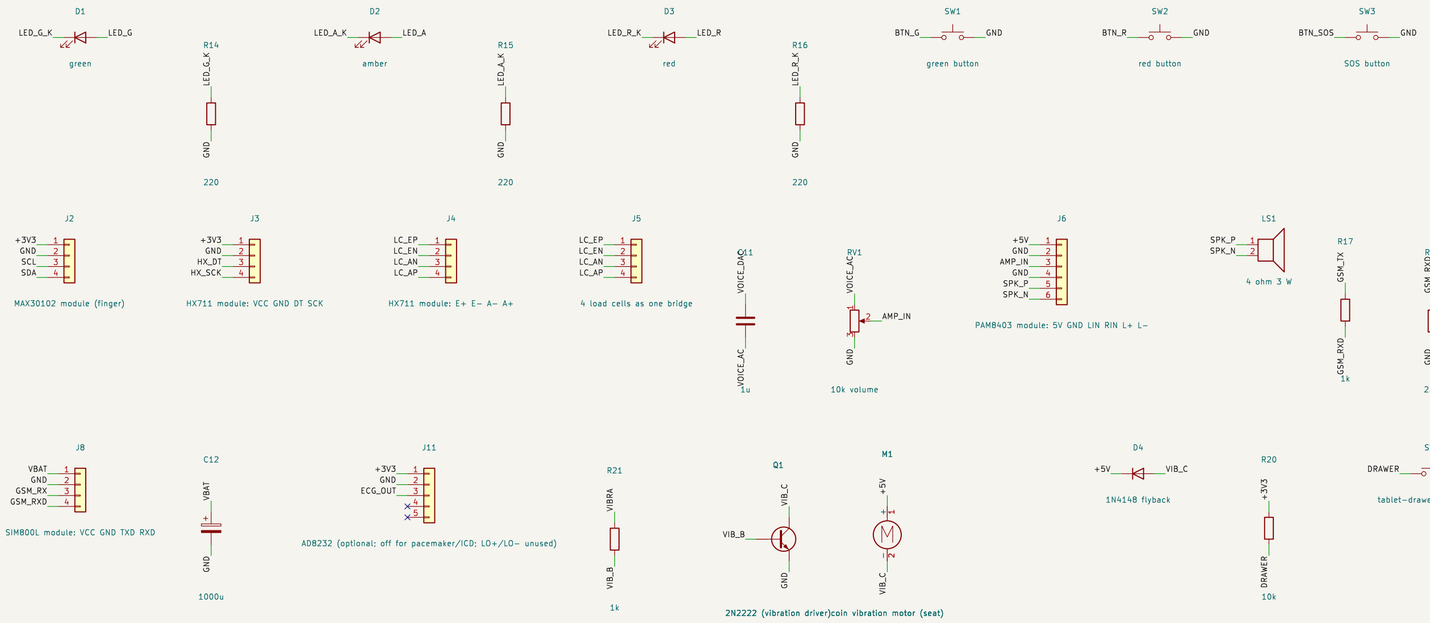
Impedance front end (custom circuit)



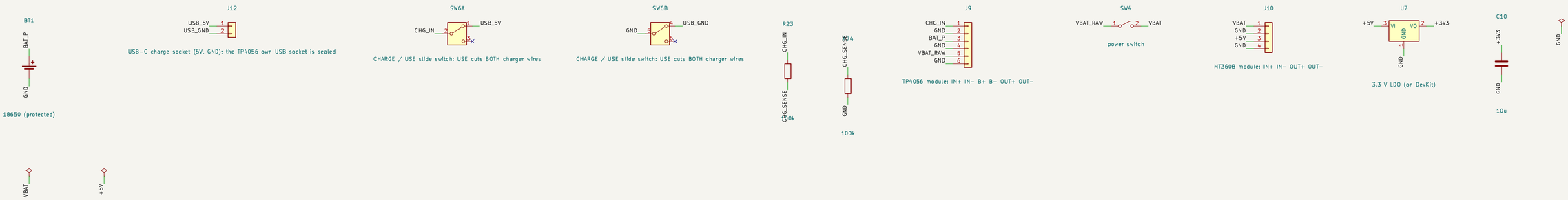
ESP32 DevKit (WROOM-32)



Sensors, user interface, SMS



Power (battery only while measuring)



Monitoring, not diagnosis. Battery power only while a person touches the electrodes; never test on anyone with a pacemaker or ICD.

Z = R5 x V(body) / V(R5): the gain cancels. Paths (S1 S0): 00 legs, 01 arms, 10 right side, 11 left side. Trunk = (Z3 + Z4 − Z1 − Z2) / 2.

R8/R9 = 10 Mohm (LTspice Monte Carlo: 1 Mohm gave up to 12 % error). Firmware reads the 5 kHz Goertzel amplitude, never peak-to-peak (mains hum).

Pin-compatible symbols: INA128 symbol used for INA126, CD4052B/CD4053B for 74HC4052/74HC4053, LM2904 for MCP6002.

Monitoring, not diagnosis
Generated by tools/make_schematic.py from the same values as the LTspice netlists
Transfinitte hardware track

Sheet: /
File: hridaymitra.kicad_sch

Title: HridayMitra stool: impedance front end + ESP32

Size: User	Date: 2026-10-24	**Rev: v3.1 (R8/R9 10M)**
KiCad E.D.A. 9.0.7		Id: 1/1